

GAIA BIANCIOTTO

Data Science • Bioinformatics • Computational Genomics

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PROFILE

Data Science and Artificial Intelligence graduate with research experience in **computational genomics**, **machine learning** and **probabilistic modelling**. Currently conducting research on CRISPR–Cas9 off-target prediction at Généthon and pursuing the M2 GENIOMHE-AI at Université Paris-Saclay. Interested in bioinformatics, genomic data analysis, precision medicine and computational methods for biomedical research.

EDUCATION

Université Paris-Saclay – Évry, France **2026 – Present**
M2 Genomics, Informatics, Mathematics and Artificial Intelligence for Health and Environment (GENIOMHE-AI)
Advanced training in bioinformatics, genomics, artificial intelligence and precision medicine.

DSTI School of Engineering – Paris, France **2024–2026**
MSc Data Science and Artificial Intelligence
Two-year Master's programme in data science, artificial intelligence and computer science.

Institut Polytechnique de Paris & HEC Paris – Paris, France **2025**
Hi! PARIS Summer School – Artificial Intelligence and Data Science

University of Bologna – Bologna, Italy **2022–2024**
Bachelor's Degree in Political and Social Sciences
Completed in two years instead of three, with a focus on quantitative and statistical methods.

RESEARCH & PROFESSIONAL EXPERIENCE

Généthon / Inserm – Évry, France **July–September 2026**
Software Engineering Assistant – Full-time Contract
Continued the development and validation of computational methods for **CRISPR–Cas9 off-target prediction**, including genome-wide analysis, benchmarking and maintenance of reproducible Python pipelines.

Généthon – Évry, France **January–June 2026**
Research Intern – Computational Genomics
Developed and evaluated probabilistic methods for **CRISPR–Cas9 off-target prediction** using large-scale genomic and experimental datasets, with a focus on sequence alignment, candidate detection and uncertainty modelling.

HEC Paris – Department of Economics and Decision Sciences **October–December 2025**
Research Assistant – ION Management Science Lab
Developed multimodal data pipelines transforming video and audio recordings into structured datasets for quantitative research on mentorship dynamics and AI-driven experimentation.

SELECTED PRESENTATION

Symposium on Generative AI for Biochemistry and Health **May 2026**
Poster Presentation – “Probabilistic modelling approaches for improving CRISPR off-target prediction by incorporating alignment uncertainty”, 4th of May 2026 at Télécom Paris

TECHNICAL SKILLS

Programming & Scripting: Python, R, SQL, Bash/Linux, PowerShell, C/C++

Machine Learning & Statistics: scikit-learn, PyTorch, TensorFlow, deep learning, statistical and probabilistic modelling, uncertainty analysis, model evaluation and experimentation; *currently developing through M2 GENIOMHE-AI*: advanced deep learning, Transformers, generative models, reinforcement learning, Bayesian inference and statistical learning

Bioinformatics & Genomics: CRISPR–Cas9 off-target analysis, genomic sequence analysis, sequence alignment, HMM/PHMM, genome-wide search, GUIDE-seq and CHANGE-seq analysis; *currently developing through M2 GENIOMHE-AI*: RNA-seq and gene expression analysis, statistical genomics, Bayesian networks, gene regulatory network analysis, systems and network biology, structural bioinformatics, RNA structure analysis and precision medicine genomics

Data Engineering & Big Data: Data pipelines, Spark, Hadoop, AWS, cloud computing, large-scale data processing and data engineering

Software & Tools: Git, Docker, MLOps, Neo4j, MongoDB, reproducible computational pipelines, version control and software development

Research & Evaluation: Model benchmarking, ROC–AUC, precision–recall analysis, experimental validation, quantitative analysis and reproducible computational research

LANGUAGES & CERTIFICATIONS

Languages: Italian (Native) • English (C1) • French (B2) • German (Basic)

Certification: Neo4j Certified Professional